be $8\ km$, while the distance to the western most point from P to be $6\ km$. If the angle between the two lines of sight is $60°$, find the width of the pond.

9. Two Navy helicopters A and B are flying over the Bay of Bengal at same altitude from the sea level to search a missing boat. Pilots of both the helicopters sight the boat at the same time while they are apart $10\ km$ from each other. If the distance of the boat from A is $6\ km$ and if the line segment AB subtends $60°$ at the boat, find the distance of the boat from B.
10. A straight tunnel is to be made through a mountain. A surveyor observes the two extremities A and B of the tunnel to be built from a point P in front of the mountain. If $AP = 3km$, $BP = 5\ km$ and $\angle APB = 120°$, then find the length of the tunnel to be built.
11. A farmer wants to purchase a triangular shaped land with sides $120 feet$ and $60 feet$ and the angle included between these two sides is $60°$. If the land costs ₹500 per sq.ft, find the amount he needed to purchase the land. Also find the perimeter of the land.
12. A fighter jet has to hit a small target by flying a horizontal distance. When the target is sighted, the pilot measures the angle of depression to be $30°$. If after $100\ km$, the target has an angle of depression of $60°$, how far is the target from the fighter jet at that instant?
13. A plane is $1\ km$ from one landmark and $2\ km$ from another. From the planes point of view the land between them subtends an angle of $45°$. How far apart are the landmarks?
14. A man starts his morning walk at a point A reaches two points B and C and finally back to A such that $\angle A = 60°$ and $\angle B = 45°,\ AC = 4km$ in the $\triangle ABC$. Find the total distance he covered during his morning walk.
15. Two vehicles leave the same place P at the same time moving along two different roads. One vehicle moves at an average speed of $60km/hr$ and the other vehicle moves at an average speed of $80\ km/hr$. After half an hour the vehicle reach the destinations A and B. If AB subtends $60°$ at the initial point P, then find AB.
16. Suppose that a satellite in space, an earth station and the centre of earth all lie in the same plane. Let r be the radius of earth and R be the distance from the centre of earth to the satellite. Let d be the distance from the earth station to the satellite. Let $30°$ be the angle of elevation from the earth station to the satellite. If the line segment connecting earth station and satellite subtends angle α at the centre of earth, then prove that $d = R\sqrt{1 + \left(\frac{r}{R}\right)^2 - 2\frac{r}{R}\cos\alpha}$.

3.9 Inverse Trigonometric Functions

A function $f(x)$ has inverse if and only if it is one-to-one and onto. Thus, inverse of a function cannot be defined if it fails to be one-to-one. However, if we restrict the domain suitably, we can make the function to be one-to-one in the restricted domain. For example, $y = x^2$ is not one-to-one for all real numbers. But $y = x^2$ is one-to-one and onto either for $x \geq 0$ or $x \leq 0$. Hence $y = x^2, x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}, x \geq 0$. Now, owing to their periodicity, none of six trigonometric functions is one-to-one over their natural domains. We shall restrict their domains so that trigonometric functions are one-to-one enabling the existence of their inverse functions. This restriction can be done in many ways once again due to their periodicity. The conventional choices for the restricted domains are arbitrary but they have some important characteristics. Each restricted domain includes the number 0 and some positive angles and the image of restricted domain contains the entire range.

Let us define the inverse of sine function. Consider $f(x) = \sin x, x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then, $\sin x$ is one-to-one and onto in this restricted domain. Hence, the inverse of sine function exists. Note that $f^{-1}(y) = x$ if and only if $f(x) = y$. We write $f^{-1}(x) = \sin^{-1}(x)$. Thus, inverse of sine is defined as $\sin^{-1}(y) = x$ if and only if $\sin x = y$.

Clearly, $\sin x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ and $\sin^{-1} x : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Thus, $\sin^{-1} t$ is an angle whose sine is equal to t and which is located in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Similarly we can define the other inverse trigonometric functions.